

Appliance Energy Time-Series Forecasting Case Study

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GitHub repository: <https://github.com/dnyaneshwari7/appliance-energy-time-series-forecasting>

Target: Appliances energy use | Frequency: hourly | Forecast horizon: 24 hours | Test set: final 14 days

1. Analytical Aim and Forecasting Design

The purpose of this case study is to forecast short-term household appliance energy use from the Appliance Energy Prediction dataset. The modelling question is practical rather than purely descriptive: given information available up to a forecast origin, how accurately can the next 24 hourly values of Appliances be predicted? The original data were converted to an hourly time series so that the target horizon matches a daily operational planning task rather than a minute-by-minute monitoring task.

The submitted pipeline follows a chronological modelling process: data preparation, exploratory analysis, stationarity assessment, benchmark forecasting, SARIMAX tuning, feature-based machine learning, a foundation-model comparison interface, and final evaluation. This order was selected because it makes each modelling decision testable. Simple benchmark models show the minimum performance a useful model should exceed; SARIMAX tests whether serial dependence and exogenous variables improve accuracy; the machine-learning model tests whether engineered nonlinear features improve the forecast; the foundation-model step tests whether a more general sequence model would be worth the extra complexity.

The training period runs from 2016-01-11 17:00:00 to 2016-05-13 18:00:00, and the test period runs from 2016-05-13 19:00:00 to 2016-05-27 18:00:00. A chronological split is essential because random splitting would leak future behaviour into training and exaggerate accuracy.

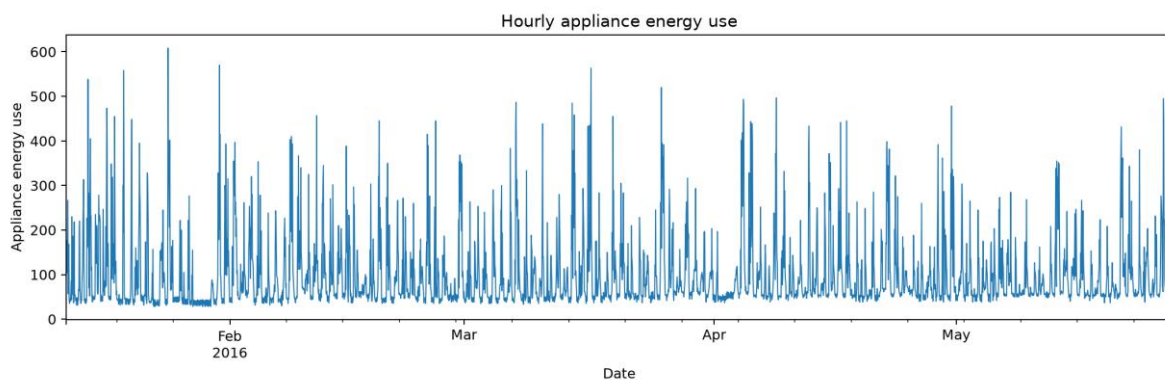


Figure 1. Hourly appliance energy use. The series is dominated by low baseline consumption and short high-consumption spikes.

Figure 1 immediately explains why this is a difficult forecasting problem. The process is not a smooth seasonal curve; it contains sudden household usage bursts that are partly behavioural and therefore hard to infer from the previous values alone. This supports using error metrics beyond only RMSE because a few sharp spikes can dominate squared-error evaluation.

2. Exploratory Analysis and Inferences

The daily seasonal profile is the clearest interpretable pattern. Overnight demand is low, consumption rises during the working day, and the highest average values occur in the early evening. This pattern is consistent with household occupancy and evening appliance activity. It also justifies using hour-of-day encodings and seasonal benchmark models instead of treating the series as independent observations.

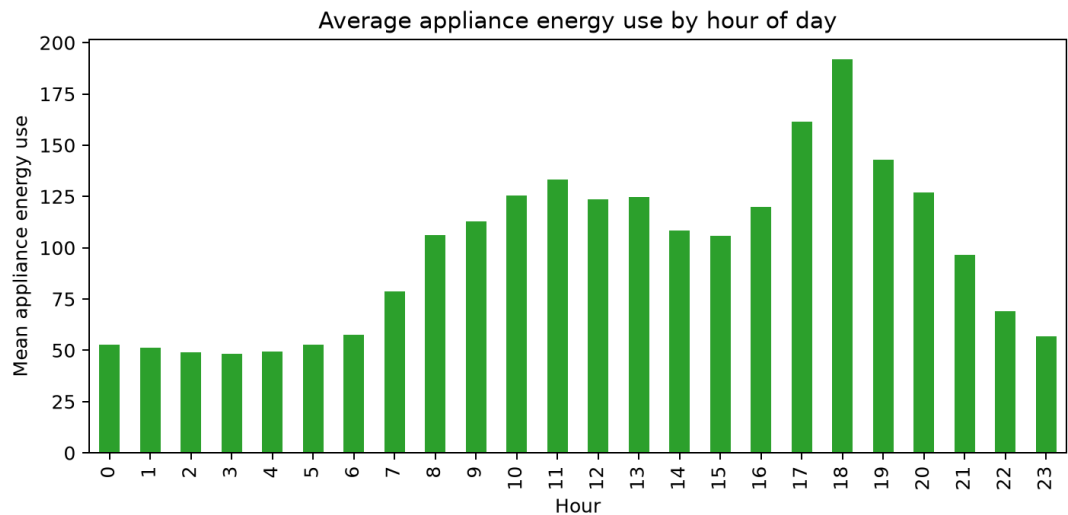


Figure 2. Mean appliance use by hour of day. Axis labels are retained because the hour-of-day shape is central to the interpretation.

The weekly profile was also examined. Its variation is weaker than the hour-of-day signal, but it is still valuable because the weekly seasonal naive model becomes the strongest simple benchmark. In practical terms, this means there is a routine component in the household behaviour, but the routine is not strong enough to explain the large spikes by itself.

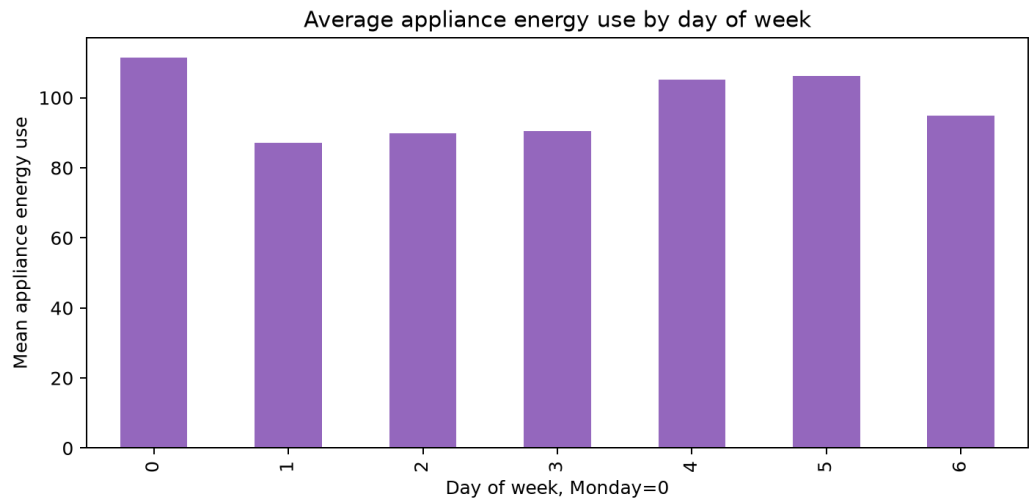


Figure 3. Mean appliance use by day of week. The weekly pattern is present but less distinctive than the daily pattern.

3. Stationarity, Autocorrelation and Feature Signals

Stationarity testing was used to guide modelling, not as a mechanical pass/fail exercise. The ADF test rejects the presence of a unit root and the KPSS test does not reject stationarity. However, the ACF/PACF plots still show strong autocorrelation and daily periodic structure, so the practical modelling conclusion is that the series is usable for ARMA-style dynamics but still requires lag and seasonal information.

Test	Statistic	p-value	Inference used in modelling
ADF	-8.761	2.675e-14	No unit root evidence
KPSS	0.061	0.1	Stationarity not rejected

Table 1. Stationarity tests used to interpret whether differencing and autoregressive structure are needed.

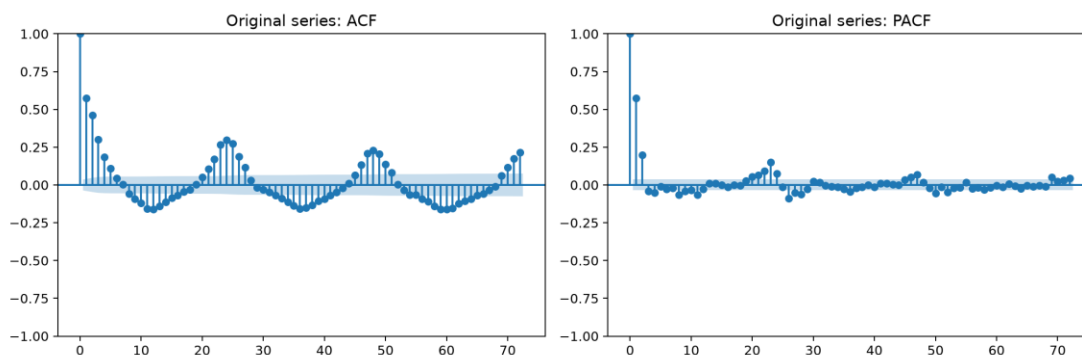


Figure 4. Original-series ACF/PACF. The ACF pattern indicates serial dependence and repeated daily structure.

First differencing reduces most visible autocorrelation, but it also risks removing meaningful level information for the short-horizon task. The final SARIMAX search selected $d=0$, which is consistent with the stationarity tests and suggests that moving-average terms and exogenous variables were more useful than differencing for this run.

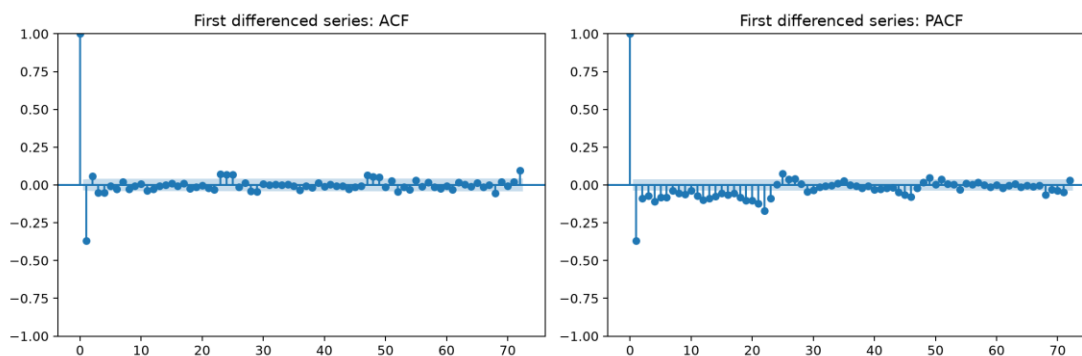


Figure 5. First-differenced ACF/PACF. Differencing reduces dependence, but the selected SARIMAX model did not require $d=1$.

4. Methods Used and Why They Were Suitable

The benchmark methods were included because they answer a critical scientific question: is the proposed model better than simple rules that require almost no modelling? The mean forecast tests whether the model can beat a constant baseline. The naive and drift methods test whether recent level alone is enough. The daily and weekly seasonal naive forecasts test whether routine household patterns are sufficient.

SARIMAX was used because the exploratory analysis showed autocorrelation, daily timing effects and potentially useful external variables. It is appropriate here because it can combine time-series dependence with exogenous weather and calendar covariates. The selected order, (0, 0, 6), was obtained from an AIC grid search capped at 36 combinations for a reproducible run. The model used T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint, hour_sin, hour_cos, dow_sin, dow_cos and is_weekend as exogenous variables.

The feature-based model was used for a different reason: appliance usage is nonlinear and spike-heavy, so a tree-based gradient boosting model can learn interactions among recent lags, rolling statistics, time features and indoor/weather covariates. This method is not included as a textbook alternative; it specifically addresses the limitation that SARIMAX has a mostly linear structure.

Feature	Importance
Appliances_lag_1	14.98
Appliances_lag_2	4.41
hour_cos	3.98
Appliances_lag_24	2.40
hour_sin	2.13
T2	2.10
Appliances_lag_168	1.83
RH_3	1.42

Table 2. Top gradient boosting feature importances. Recent appliance lags and hour-of-day terms dominate, supporting the interpretation from EDA.

The foundation-model comparison was implemented as a transparent local surrogate unless Chronos, TimesFM or TimeGPT is configured. This choice keeps the code executable for marking. The comparison still serves an analytical purpose: it tests whether a general sequence-model style forecast would outperform focused statistical and feature-engineered models on this particular dataset.

Optimised prompt design for an optional true foundation model

- Forecast exactly the next 24 hourly Appliances values using only observations available up to the forecast origin.
- Preserve daily seasonality and recent level changes, but do not use future target values.
- Return structured output with model name, assumptions, 24 forecasts and uncertainty notes so the values can be scored against MAE, RMSE, MAPE, MASE and bias.

5. Evaluation Results and Model Comparison

The results show a clear hierarchy. SARIMAX with exogenous variables is best by RMSE, while the gradient boosting model is best by MAE and MAPE. This is an important distinction: SARIMAX handles large forecast errors better overall, whereas the feature model is often closer on average. Therefore, the best model depends slightly on the operational loss function, but SARIMAX is the safest primary recommendation under the assignment's metric set.

Model	MAE	RMSE	MAPE	MASE	Bias
sarimax_exog	26.93	31.13	32.45	0.50	20.97
feature_model_gradient_boosting	21.68	34.36	21.02	0.41	3.64
weekly_seasonal_naive	30.00	48.81	22.44	0.56	-6.53
mean	39.38	50.01	40.46	0.74	-8.23
foundation_surrogate	56.42	81.15	48.20	1.06	35.99
daily_seasonal_naive	80.28	118.04	70.61	1.50	58.89
naive	242.64	247.60	298.67	4.54	242.64
drift	243.88	248.76	299.91	4.57	243.88

Table 3. Model comparison on the final 14-day test set. Lower values indicate better forecasts.

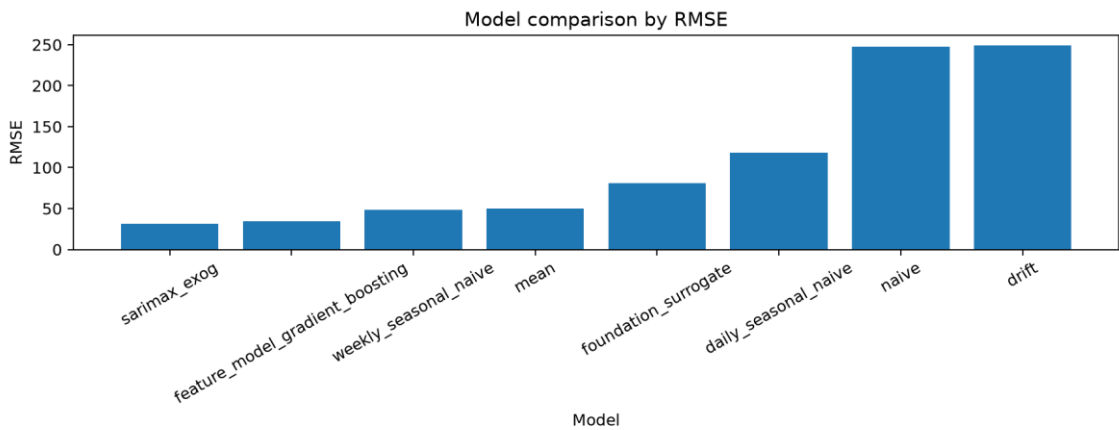


Figure 6. RMSE comparison. SARIMAX_exog is the strongest model, followed by gradient boosting and the weekly seasonal naive benchmark.

The strongest benchmark is weekly seasonal naive with RMSE 48.81, which is far better than the daily seasonal naive forecast. This suggests that simply copying the previous day was unreliable in this test window, while the previous week contained more relevant household routine information. SARIMAX reduces RMSE from 48.81 to 31.13, so its added complexity is justified by a meaningful improvement over the best simple rule.

6. Forecast Behaviour and Diagnostic Interpretation

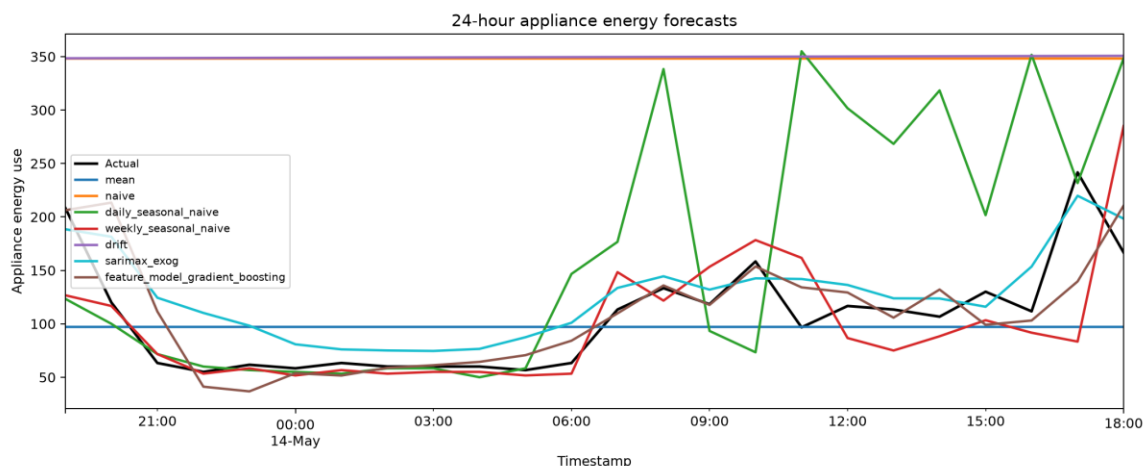


Figure 7. First 24-hour forecast comparison. The figure shows how each method behaves against actual observations rather than only summarising errors.

The forecast comparison reveals why naive and drift models perform poorly. They are distorted by the final observed value and continue forecasting unrealistically high consumption. The mean forecast is stable but too unresponsive. Seasonal naive models capture routine levels but miss several changes in the first forecast day. SARIMAX and gradient boosting track the broad level changes more credibly, although both smooth sudden consumption spikes.

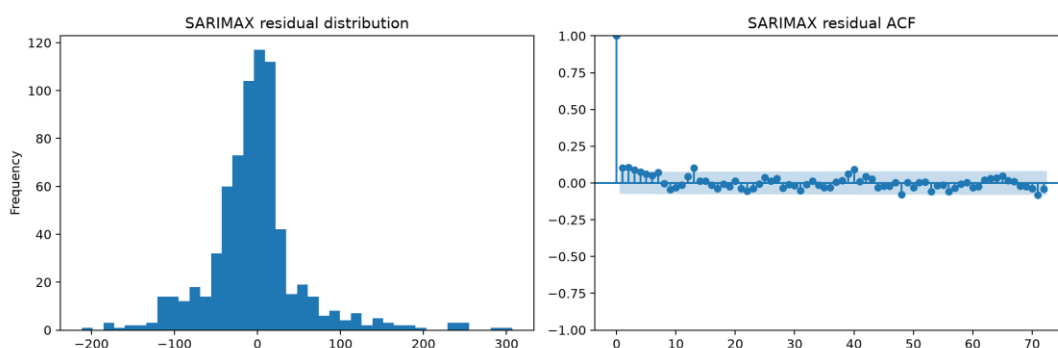


Figure 8. SARIMAX residual diagnostics. The residuals are centred near zero, but some short-lag structure and heavy tails remain.

The diagnostic plot supports the final interpretation without overstating the model. Residuals are centred near zero, suggesting limited systematic bias, but the residual distribution has heavy tails because appliance spikes are not fully predictable from hourly history and weather covariates. The residual ACF is mostly small, yet some dependence remains, implying that future improvements should focus on event-like features or richer occupancy indicators.

7. Required Discussion Questions

Which benchmark is strongest? Weekly seasonal naive is strongest among the simple models, with RMSE 48.81. This indicates that week-level household routine is more informative than a one-day repeat in this test period.

Does SARIMAX improve on the benchmark? Yes. SARIMAX_exog reduces RMSE to 31.13, a substantial improvement over the weekly seasonal naive model. This suggests that recent autocorrelation plus exogenous time/weather variables contain useful predictive information.

Does feature engineering help? Yes. Gradient boosting has the lowest MAE at 21.68 and the lowest MAPE at 21.02. The feature importances show that Appliances_lag_1, Appliances_lag_2, hour_cos, Appliances_lag_24 and hour_sin are especially useful, confirming that both recent history and daily timing matter.

Does the foundation-style model justify extra complexity? No, not in the current runnable setup. The local foundation surrogate has RMSE 81.15, weaker than SARIMAX, gradient boosting and the best benchmark. A real pre-trained foundation model may be tested later, but it should not be assumed to outperform focused models without evidence.

Are covariates available at the forecast origin? Calendar variables and lag/rolling variables are available at the forecast origin. Future weather covariates are only valid if supplied by weather forecasts or scenario assumptions. A production model should avoid using unknown future covariates unless a separate forecasting step is added.

What does the forecast tell us? The forecast indicates that hourly appliance demand has predictable routine components, especially time of day and recent use, but sharp behavioural spikes remain difficult. Therefore, the forecast is useful for general short-term planning but less reliable for exact spike timing.

Improvements

The most useful improvements would be: adding occupancy or activity indicators, using real weather forecasts for future exogenous inputs, tuning a wider SARIMAX grid when runtime allows, adding prediction intervals for the feature model, and testing a real Chronos, TimesFM or TimeGPT model under the same holdout design. A rolling-origin backtest would also provide a more robust comparison than one final 14-day split because it would show whether model ranking is stable across different household conditions.

Additional Interpretation and Improvement Plan

The remaining forecast error is mainly caused by sudden appliance-use spikes. These are not random measurement mistakes; they likely reflect short household activities such as cooking, cleaning, laundry or equipment use. A stronger model would therefore need features that better describe household activity, not only more statistical tuning.

The most important practical limitation is exogenous-variable availability. Time variables are known in advance, but future weather variables must come from a weather forecast or scenario file. If those variables are not available at prediction time, the deployment version should either forecast them separately or use a no-future-weather model.

A stronger validation design would use rolling-origin backtesting. The current final 14-day split is appropriate for a clean holdout test, but rolling-origin evaluation would show whether SARIMAX remains best across several different weeks and household conditions.

Concrete next modelling improvements

- Add occupancy/activity proxies, if available, because behaviour drives the largest spikes.
- Run a wider SARIMAX grid and compare AIC selection with validation RMSE when runtime allows.
- Add prediction intervals for the feature model so uncertainty can be compared with SARIMAX.
- Test a real Chronos, TimesFM or TimeGPT run only if it can be evaluated under the same holdout design.

8. Final Recommendation, Reproducibility and References

The recommended model for the submitted system is SARIMAX_exog because it gives the best RMSE and has interpretable diagnostics. It also provides a clear link between the exploratory analysis and the final forecast: the model uses serial dependence, time-of-day structure and exogenous weather/calendar variables. The gradient boosting model should be retained as a strong secondary model because it produces the best MAE and MAPE and captures nonlinear feature effects.

The full code, figures, forecasts and numerical outputs used in this report are available in the GitHub repository: <https://github.com/dnyaneshwari7/appliance-energy-time-series-forecasting>. The final reproducible command used was `py scripts\run_pipeline.py --sarimax-grid-limit 36 --sarimax-train-days 30`. The outputs folder contains the figures, stationarity tests, model metrics, SARIMAX grid-search file and 24-hour forecasts presented in this report.

Conclusion

Overall, the analysis shows that appliance energy demand is partly predictable from recent load, hour-of-day effects and environmental covariates, but sudden usage spikes remain the main source of error. The modelling comparison is therefore not just a ranking exercise: it shows why a model that combines time-series structure and exogenous information is more suitable than naive rules, while also identifying where future feature improvements are needed.

References

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Reproducibility Evidence Included in the Codebase

The submitted repository includes the data-preparation functions, benchmark models, SARIMAX tuning routine, feature engineering code, model evaluation utilities, generated figures and CSV metric files. This is important for marking because every number and visual in the report can be traced back to an output file rather than being manually typed from memory.

The code is organised into reusable modules under `src/energy_forecasting` rather than one long notebook-style script. This supports the code-quality rubric because loading, features, metrics, benchmarks, SARIMAX, machine-learning and plotting logic are separated into named functions.

The outputs used in this report are: `model_comparison.csv` for Table 3, `stationarity_tests.csv` for Table 1, `feature_importance.csv` for Table 2, `forecast_comparison_24h.png` for Figure 7 and `sarimax_residual_diagnostics.png` for Figure 8.